

CTE Assignment (50 questions)

#1 Find total sales per customer using cte

with cust_sales as

```
(select c.customer_name,sum(s.total_amount) as total_amt from customers c join sales s
using(customer_id)
```

```
group by c.customer_id,c.customer_name)
```

```
select customer_name,total_amt from cust_sales order by total_amt desc;
```

#2 Find customers whose total sales are above average using cte

with cust_sales as

```
(select c.customer_name,sum(s.total_amount) as total_amt from customers c join sales s
using(customer_id)
```

```
group by c.customer_id,c.customer_name),
```

cust_avg as

```
(select avg(total_amt) as avg_amt from cust_sales)
```

```
select customer_name,total_amt from cust_sales where total_amt> (select avg_amt from
cust_avg);
```

#3 Find products with total quantity sold using cte

with prod_sold as

```
(select p.product_name,sum(si.quantity) as total_qty from products p join sale_items si
using(product_id)
```

```
group by p.product_id,p.product_name)
```

```
select product_name,total_qty from prod_sold order by total_qty desc;
```

#4 Find products sold more than average quantity using cte

with prod_sales as

```
(select p.product_name,sum(si.quantity) as total_qty from products p join sale_items si
using(product_id)
```

```
group by p.product_id,p.product_name),
```

prod_avg as

```
(select avg(total_qty) as avg_qty from prod_sales)
```

```
select product_name,total_qty from prod_sales where total_qty> (select avg_qty from prod_avg);
```

#5 Find employees with total sales handled using cte

with emp_sales as

```
(select e.employee_name,sum(s.total_amount) as total_amt from employees e join sales s  
using(employee_id)
```

```
group by e.employee_id,e.employee_name)
```

```
select employee_name,total_amt from emp_sales order by total_amt desc;
```

#6 Find employees who handled above average sales using cte

with emp_sales as

```
(select e.employee_name,sum(s.total_amount) as total_amt from employees e join sales s  
using(employee_id)
```

```
group by e.employee_id,e.employee_name),
```

emp_avg as

```
(select avg(total_amt) as avg_sales from emp_sales)
```

```
select employee_name,total_amt from emp_sales where total_amt> (select avg_sales from  
emp_avg);
```

#7 Find customers who spent the most using cte

with cust_sales as

```
(select c.customer_name,sum(s.total_amount) as total_amt from customers c join sales s  
using(customer_id)
```

```
group by c.customer_id,c.customer_name)
```

```
select customer_name,total_amt from cust_sales order by total_amt desc;
```

#8 Find products with highest total quantity sold using cte

with prod_sold as

```
(select p.product_name,sum(si.quantity) as total_qty from products p join sale_items si  
using(product_id)
```

```
group by p.product_id,p.product_name)

select product_name from prod_sold where total_qty=

(select max(total_qty) from prod_sold);
```

#9 Find employees with highest number of sales using cte

```
with emp_sales as

(select e.employee_name,sum(s.total_amount) as total_amt from employees e join sales s
using(employee_id)

group by e.employee_id,e.employee_name),

max_sales as

(select max(total_amt) as max_amount from emp_sales)

select employee_name from emp_sales where total_amt=

(select max_amount from max_sales);
```

#10 Find customers with lowest total spending using cte

```
with cust_sales as

(select c.customer_name,sum(s.total_amount) as total_amt from customers c join sales s
using(customer_id)

group by c.customer_id,c.customer_name),

cust_min as

(select min(total_amt) as min_amt from cust_sales)

select customer_name from cust_sales where total_amt=

(select min_amt from cust_min);
```

#11 Find products with price above average using cte

```
with prod_avg as

(select avg(price) as avg_price from products)

select product_name from products where price>(select avg_price from prod_avg);
```

#12 Find employees earning above average salary using cte

with avg_sal as

(select avg(salary) as avg_salary from employees)

select employee_name from employees where salary>(select avg_salary from avg_sal);

#13 Find customers with total orders count using cte

with cust_order as

(select customer_id,count(customer_id) as count_cust from sales group by customer_id)

select customer_name,count_cust from customers join cust_order using(customer_id);

#14 Find customers with more than 2 orders using cte

with cust_order as

(select customer_id,count(customer_id) as count_cust from sales group by customer_id)

select customer_name,count_cust from customers join cust_order using(customer_id) where count_cust>2;

#15 Find products that were never sold using cte

with not_sold as

(select distinct product_id from sale_items)

select p.product_name from products p where not exists (select 1 from not_sold ns where ns.product_id=p.product_id);

#16 Find customers who never made a purchase using cte

with not_sale as

(select distinct customer_id from sales)

select c.customer_name from customers c where not exists (select ns.customer_id from not_sale ns where c.customer_id=ns.customer_id);

#17 Find products that were returned using cte

with prod_return as

(select distinct product_id from returns)

```
select p.product_name from products p where exists(select 1 from prod_return pr where
p.product_id=pr.product_id);
```

#18 Find products that were never returned using cte

with prod_return as

(select distinct product_id from returns)

```
select p.product_name from products p where not exists(select 1 from prod_return pr where
p.product_id=pr.product_id);
```

#19 Find suppliers who supplied products using cte

with sup_prod as

(select supplier_id from products)

```
select s.supplier_name from suppliers s where exists(select 1 from sup_prod sp where
sp.supplier_id=s.supplier_id);
```

#20 Find suppliers who never supplied products using cte

with sup_prod as

(select supplier_id from products)

```
select s.supplier_name from suppliers s where not exists(select 1 from sup_prod sp where
sp.supplier_id=s.supplier_id);
```

#21 Find total quantity sold per category using cte

with cat_sold as

```
(select c.category_id,category_name,sum(si.quantity) as total_qty from categories c join
products p using(category_id)
```

```
join sale_items si using(product_id) group by c.category_id,category_name)
```

```
select category_name,total_qty from cat_sold order by total_qty desc;
```

#22 Find categories with highest sales using cte

with cat_sold as

```
(select c.category_id,category_name,sum(si.price_at_sale) as total_amt from categories c join
products p using(category_id)
```

```
join sale_items si using(product_id) group by c.category_id,category_name),  
max_cat as  
(select max(total_amt) as max_amt from cat_sold)  
select category_name,total_amt from cat_sold where total_amt=(select max_amt from  
max_cat);
```

#23 Find categories with lowest sales using cte

```
with cat_sold as  
(select c.category_id,category_name,sum(si.price_at_sale) as total_amt from categories c join  
products p using(category_id)  
join sale_items si using(product_id) group by c.category_id,category_name),  
min_cat as  
(select min(total_amt) as min_amt from cat_sold)  
select category_name,total_amt from cat_sold where total_amt=(select min_amt from  
min_cat);
```

#24 Find total revenue per product using cte

```
with prod_price as  
(select product_id,sum(price_at_sale) as total_amt from sale_items si group by product_id)  
select p.product_name,pp.total_amt from products p join prod_price pp using(product_id) order  
by pp.total_amt desc;
```

#25 Find products generating above average revenue using cte

```
with prod_price as  
(select product_id,sum(price_at_sale) as total_amt from sale_items si group by product_id),  
avg_prod as  
(select avg(total_amt) as avg_amt from prod_price)  
select p.product_name,pp.total_amt from products p join prod_price pp using(product_id)  
where pp.total_amt>  
(select avg_amt from avg_prod);
```

#26 Find employees who handled high value sales using cte

with emp_sales as

(select employee_id,sum(total_amount) as total_amt from sales group by employee_id),

max_emp as

(select max(total_amt) as max_amt from emp_sales)

select e.employee_name from employees e join emp_sales es using(employee_id) where
es.total_amt=

(select max_amt from max_emp);

#27 Find customers who bought expensive products using cte

with cust_sales as

(select customer_id,sum(total_amount) as total_amt from sales group by customer_id),

max_cust as

(select max(total_amt) as max_amt from cust_sales)

select c.customer_name from customers c join cust_sales cs using(customer_id) where
cs.total_amt=

(select max_amt from max_cust);

#28 Find products priced above maximum price of another category using cte

with cat_amt as

(select category_id,max(price) as max_price from products group by category_id)

select p.product_name,p.price from products p where p.price> any

(select ca.max_price from cat_amt ca where ca.category_id <> p.category_id);

#29 Find customers who bought more than average number of items using cte

with cust_order as

(select customer_id,count(customer_id) as no_of_cust from sales group by customer_id),

cust_avg as

(select avg(no_of_cust) as avg_cust from cust_order)

select c.customer_name from customers c join cust_order co using(customer_id) where
co.no_of_cust>

```
(select avg_cust from cust_avg);
```

#30 Find employees with less than average sales using cte

```
with emp_sales as
```

```
(select employee_id,sum(total_amount) as total_amt from sales group by employee_id),
```

```
emp_avg as
```

```
(select avg(total_amt) as avg_amt from emp_sales)
```

```
select e.employee_name from employees e join emp_sales es using(employee_id) where  
es.total_amt<
```

```
(select avg_amt from emp_avg);
```

#31 Find products with stock above average using cte

```
with stock_avg as
```

```
(select avg(stock_quantity) as avg_qty from products)
```

```
select product_name from products where stock_quantity>
```

```
(select avg_qty from stock_avg);
```

#32 Find products with stock below average using cte

```
with stock_avg as
```

```
(select avg(stock_quantity) as avg_qty from products)
```

```
select product_name from products where stock_quantity<
```

```
(select avg_qty from stock_avg);
```

#33 Find customers grouped by country with total spending using cte

```
with cust_amt as
```

```
(select c.country,sum(s.total_amount) as total_amt from customers c join sales s  
using(customer_id)
```

```
group by c.country)
```

```
select country,total_amt from cust_amt;
```


#34 Find countries with more than 3 customers using cte

with country_count as

(select country,count(country) as country_count from customers group by country having count(country)>3)

select country from country_count;

#35 Find employees grouped by role with average salary using cte

with emp_sal as

(select role,avg(salary) as avg_salary from employees group by role)

select role,avg_salary from emp_sal;

#36 Find roles with highest average salary using cte

with emp_sal as

(select role,avg(salary) as avg_salary,dense_rank()over(order by avg(salary) desc) as rank_no from employees group by role)

select role,avg_salary from emp_sal where rank_no=1;

#37 Find roles with more than one employee using cte

with count_role as

(select role,count(role) as total_no from employees group by role having count(role)>1)

select role,total_no from count_role;

#38 Find customers who joined earliest using cte

with cust_info as

(select min(join_date) as min_date from customers)

select customer_name from customers where join_date=

(select min_date from cust_info);

#39 Find customers who joined latest using cte

with cust_info as

```
(select max(join_date) as max_date from customers)
select customer_name from customers where join_date=
(select max_date from cust_info);
```

#40 Find sales per day using cte

```
with sale_day as
(select sale_date,sum(total_amount) as total_amt from sales group by sale_date)
select sale_date,total_amt from sale_day;
```

#41 Find days with highest sales using cte

```
with sale_date as
(select sale_date,sum(total_amount) as total_amt from sales group by sale_date),
max_date as
(select max(total_amt) as max_amt from sale_date)
select sale_date,total_amt from sale_date where total_amt=
(select max_amt from max_date);
```

#42 Find days with lowest sales using cte

```
with sale_date as
(select sale_date,sum(total_amount) as total_amt from sales group by sale_date),
min_date as
(select min(total_amt) as min_amt from sale_date)
select sale_date,total_amt from sale_date where total_amt=
(select min_amt from min_date);
```

#43 Find products with multiple sales entries using cte

```
with prod_count as
(select product_id,count(product_id) as count_product from sale_items group by product_id
having count(product_id)>1)
select product_name from products join prod_count using(product_id);
```

#44 Find sales having more than one product using cte

with prod_sale as

(select sale_id, count(product_id) as count_product from sale_items group by sale_id)

select sale_id from prod_sale where count_product > 1;

#45 Find customers who returned products using cte

with cust_return as

(select customer_id from customers join sales using(customer_id) join returns using(sale_id))

select c.customer_name from customers c where exists

(select 1 from cust_return cr where c.customer_id = cr.customer_id);

#46 Find customers who never returned products using cte

with cust_return as

(select customer_id from customers join sales using(customer_id) join returns using(sale_id))

select c.customer_name from customers c where not exists

(select 1 from cust_return cr where c.customer_id = cr.customer_id);

#47 Find employees involved in returns using cte

with emp_return as

(select employee_id from employees join sales using(employee_id) join returns using(sale_id))

select e.employee_name from employees e where exists

(select 1 from emp_return er where er.employee_id = e.employee_id);

#48 Find products that were restocked using cte

with prod_stock as

(select product_id from inventory)

select p.product_name from products p where exists

(select 1 from prod_stock ps where p.product_id = ps.product_id);

#49 Find products never restocked using cte

with prod_stock as

(select product_id from inventory)

select p.product_name from products p where not exists

(select 1 from prod_stock ps where p.product_id=ps.product_id);

#50 Find top 3 highest paid employees using cte

with emp_salary as

(select employee_id,employee_name,salary,dense_rank()over(order by salary desc) as rank_no
from employees)

select employee_name,salary from emp_salary where rank_no<=3;